HAT-P-54b

R-1217

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_HATP-54_220114` · created 2026-07-31T08:26:00+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459593.7782 +/- 0.0033 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.045 +/- 0.048
Transit depth (Rp/Rs)^2	0.2 +/- 0.43 [%]
Orbital Inclination (inc)	82.39 +/- 2.41
Airmass coefficient 1 (a1)	0.401 +/- 0.02
Airmass coefficient 2 (a2)	0.9 +/- 0.29
Scatter in the residuals of the lightcurve fit is	5.09 %
Best Comparison Star	#2 - [468, 87]
Optimal Aperture	2.58
Optimal Annulus	14.02
Transit Duration (day)	0.02 +/- 0.022

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.045 ± 0.048 vs 0.1572 (deviation 1.69σ)
Duration fit vs published	0.02 d vs 0.075 d (deviation 1.81σ)
Mid-time O–C	1.1 min
Depth SNR	0.7
Residual scatter	5.09 %

Light curve

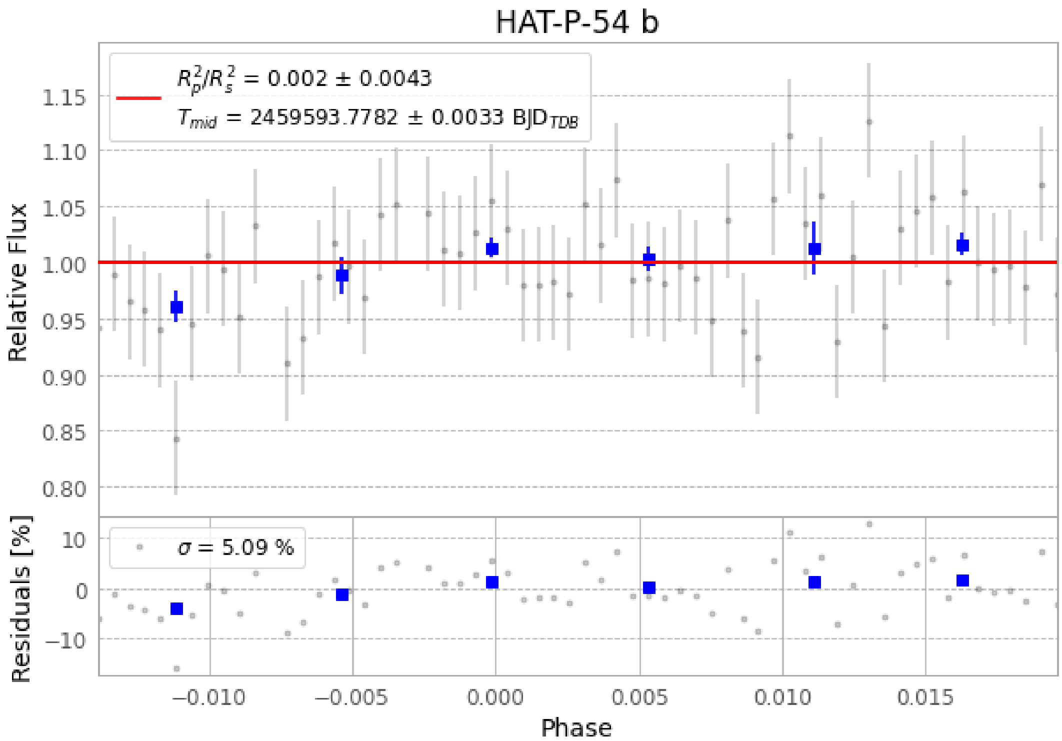

Figure 1 — FinalLightCurve_HAT-P-54 b_2022-01-13.png (SHA-256 f4918c3009fe9f6a…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [194, 230], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 e6f46e5a8e45009f…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b`

Data provenance

62 FITS frames (41 MB), `HATP-54220114051513.FITS` → `HATP-54220114081812.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-54-220114.log (d215393c3afc…), FinalLightCurve_HAT-P-54 b_2022-01-13.png (f4918c3009fe…), FinalLightCurve_HAT-P-54 b_2022-01-13.csv (fd6812040782…)

Compute resources

Wall time 135.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)